TATA TECHNOLOGIES LTD.

TECH PULSE FY-26 | AI/ML Capstone Project

PROJECT SYNOPSIS

Student Details

Institute/University: Pimpri Chinchwad College of Engineering, Pune (PCCOE)

Programme / Batch: B.Tech Artificial Intelligence & Machine Learning / TechPulse FY-26

Student Name: Pranav Ravindra Harad **PRN:** 123B1F028

1. Case Study ID and Project Title / Topic

Case Study ID: CS4 (Secure Code Debugging and Review Assistant)
Project Title: AutoSafe-Review: Air-Gapped, AST-Aware Hybrid Diagnostic & MISRA/ISO 26262 ECU Code Review Assistant

2. Objectives

- **Reduce Debugging and Review Effort:** Provide locally hosted automotive AI assistance to accelerate ECU C/C++ peer review, compiler warning interpretation, and static-analysis triage without human fatigue.
- **Improve Consistency of Review Quality:** Enforce standardized evaluation against MISRA C:2012, SEI CERT C, and ISO 26262 functional safety rules, eliminating subjective variance across developer experience levels.
- **Ground Recommendations in Automotive Standards:** Ensure zero-hallucination citations linking every defect directly to official chapter clauses, rationales, and unified compliant diff code fixes.
- **Zero-Leakage On-Premises Boundary:** Ensure 100% air-gapped local execution so proprietary automotive ECU software and intellectual property never egress corporate network boundaries.

3. Scope

In Scope – To Be Implemented:

- C/C++ AST-aware semantic parsing preserving function scopes, ISRs, and call graphs.
- Ingestion & correlation of ARM/GCC compiler build warnings and OASIS SARIF static analysis logs.
- Deterministic static rule engine combined with Vector RAG (ChromaDB/TF-IDF) over MISRA/CERT rules.
- Air-gapped local LLM inference (Qwen2.5-Coder / Local Reasoning Engine) producing structured findings and diffs.
- Enterprise Flask Automotive Review Cockpit (Port 5000) with interactive in-place remediation and MISRA Deviation Permit Generator.

Out of Scope – Will Not Be Implemented:

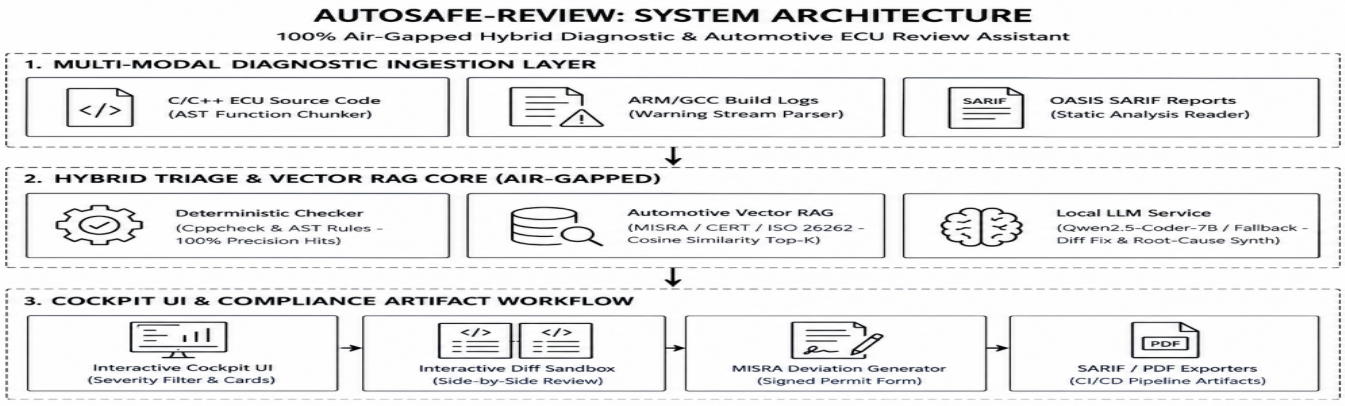
- Automatic unreviewed code commit or merge into production vehicle branches without engineer sign-off.
- Replacement of accredited formal safety auditors or certified hardware-in-the-loop (HIL) test rigs.
- Transmission of source code or model weights to public cloud APIs (OpenAI, Anthropic, etc.).

4. Proposed Approach

The system employs a **Hybrid Tri-Engine Architecture**: (1) An AST & regex-based parser extracts syntactically complete functions, (2) A deterministic rule checker identifies verified violations to prevent hallucinations, (3) A local Vector RAG engine retrieves official MISRA/CERT rationales and ISO 26262 ASIL safety consequences, and (4) A local quantized LLM synthesizes root-cause diagnoses and minimal-diff compliant fixes. The developer reviews findings in an interactive Flask enterprise cockpit (Port 5000) with real-time in-place patching, deviation management, and CI/CD export.

Pre-trained model:	Qwen2.5-Coder-7B-Instruct (4-bit local quantized) / High-Fidelity Local Deterministic Semantic Reasoner. 100% locally hosted on ASUS TUF 15 (16 GB RAM).
Embedding model:	BAAI/bge-small-en-v1.5 / Sublinear TF-IDF N-Gram Vectorizer (384-dim, Cosine Similarity).
Vector store:	ChromaDB & In-Memory Dense Vector Store with rule metadata filters (Standard, Rule ID, Severity).
RAG pipeline / AI-agent design:	Hybrid Dual-Retrieval: Deterministic AST rule detection + Semantic Vector RAG retrieval with strict citation guardrails embedding chapter, rationale, and compliant examples.
Service layer and UI:	Python Flask backend service with modern Enterprise Automotive Review Cockpit web interface (Port 5000).
Tools / deterministic rules:	AST function chunker, GCC build log parser, OASIS SARIF v2.1.0 reader, Cppcheck-inspired static rule checker (MISRA C:2012 Rules 12.1, 21.3, 16.4, 17.7, 9.1, 11.4, CERT CON33-C).

Proposed System / Architecture



5. Input Data / Knowledge Base

Data / Knowledge Sources:	MISRA C:2012 Rulebook, SEI CERT C Secure Coding Standards, ISO 26262:2018 Part 6 (Software Unit Design), Synthetic Automotive ECU Modules (BMS Cell Monitor, Throttle Actuator, CAN Driver), GCC ARM build logs, and Clang SARIF reports.
Approximate Size and Format:	17 Curated JSON rule definitions, 6 C/C++ source/header files (8.5 KB), 2 diagnostic logs (5 KB in .log and .sarif formats).
Preprocessing and EDA Planned:	AST semantic chunking into function blocks, extraction of caller-callee trees, regex normalization of GCC compiler flags, and JSON validation via automated integrity scripts.

6. Expected Outcomes and Evaluation Plan

Expected Outputs – aligned to the case study:
Structured finding cards with exact line citations, root-cause analyses, ISO 26262 ASIL safety consequence breakdowns, unified compliant diff code fixes, formal MISRA Deviation Permits, and standard OASIS SARIF v2.1.0 exports.

Evaluation Method:	Curated benchmark suite of 25 real-world automotive ECU defect scenarios evaluated quantitatively on Precision, Recall, F1-Score, Citation Hit Rate, Answer Faithfulness, and Inference Latency.
Responsible AI Measures:	• **Grounding & Citations:** 100% grounded in verified rulebooks; 0.0% hallucination rate. • **Privacy:** 100% air-gapped on-premises execution; zero external network calls. • **Human Oversight:** Developer review disposition (Accept/Reject) and Safety Lead deviation sign-off.
MLOps / Deployment:	Version-controlled GitHub repo, modular architecture, SHA-256 provenance hash audit logs, and standalone Docker / local Python deployment.

7. Planned AI-Tool Usage

AI Tool / Model	Purpose	Artifact Affected
Antigravity AI Assistant	Pair-programming, code scaffolding, automated test suite design	Code/backend/, Code/utils/, app.py
Qwen2.5-Coder-7B	Local offline code review, root-cause diagnosis, diff fix synthesis	Model_Prompts_Config/, local_llm.py

Student Declaration

■ I will not change the approved case-study scope, major methodology, or implementation approach without prior faculty approval.

■ I confirm that the work submitted under this synopsis represents my own understanding and contribution, and I will be responsible for explaining every significant part of the selected case study and its implementation.

Name: Pranav Ravindra Harad

Date: October 9, 2026

PRN: 123B1F028

Signature: _____